COURSE SYLLABUS

Linux Administration Essentials

COURSE NUMBER	DURATION	AUDIENCE LEVEL	FORMAT
LNX-001	1 day · 4 modules	Beginner	Instructor-Led / Self-Paced

Course syllabus

COURSE OVERVIEW

This one-day course will teach the basics of how to administer a Linux system running Ubuntu. Students will learn administration of user accounts, computer hardware, networking and security.

AT COURSE COMPLETION

After completing this course, you will be able to:

Identify the core concepts and administrative file paths used in standard Linux environments.

Navigate Ubuntu command-line interfaces, basic networking components, and system configurations.

Perform primary management tasks including user grouping, volume mounting, and remote-access configuration.

Apply safe execution practices and file controls in realistic administrative scenarios.

Validate service operations and inspect logs to troubleshoot issues on modern Linux systems.

PREREQUISITES

- Basic understanding of computer operating systems
- Familiarity with foundational command-line concepts

COURSE OUTLINE

MODULE 1 Linux User and Group Administration

90 min

Learn how to manage user accounts, group memberships, and standard file permissions on an Ubuntu system.

LESSONS

- **Lesson 1:** Managing User Accounts and Group Memberships
- **Lesson 2:** Understanding Linux File Ownership and Permissions
- **Lesson 3:** Sudo Architecture and Administrative Access Control

LAB EXERCISES

- Configuring Users, Groups, and Permissions on Ubuntu

AFTER COMPLETING THIS MODULE, YOU WILL BE ABLE TO

- Create and manage local user accounts and groups using terminal commands
- Configure standard Linux file permissions and ownership configurations
- Utilize administrative privileges securely with the sudo configuration framework
- Identify the core concepts and terminology used in Linux Administration Essentials

MODULE 2 Hardware and File System Management

90 min

Explore system hardware identification, disk partitioning, and filesystem mounting on Ubuntu.

LESSONS

- **Lesson 1:** Hardware Discovery and Querying Tools
- **Lesson 2:** Disk Partitioning and Filesystem Creation
- **Lesson 3:** The Mounting Mechanism and fstab Configuration

LAB EXERCISES

- Adding and Mounting a New Virtual Storage Volume

AFTER COMPLETING THIS MODULE, YOU WILL BE ABLE TO

- Identify and query system hardware information using built-in system tools
- Perform basic partition operations on available system storage devices
- Format and mount custom file systems reliably across system reboots
- Identify the core concepts and terminology used in Linux Administration Essentials

MODULE 3 Linux Network Configuration and Services

90 min

Learn how to configure network interfaces, troubleshoot connectivity, and manage administrative remote services.

LESSONS

- **Lesson 1:** Network Interface Configuration and Tools
- **Lesson 2:** Diagnostic Utilities for Network Troubleshooting
- **Lesson 3:** Configuring Secure Shell (SSH) Access

LAB EXERCISES

Configuring Interfaces and Securing SSH

AFTER COMPLETING THIS MODULE, YOU WILL BE ABLE TO

Inspect and configure active network interface properties and routes

Verify remote network connectivity using standard diagnostic utilities

Deploy and secure the OpenSSH server daemon for remote management access

Identify the core concepts and terminology used in Linux Administration Essentials

MODULE 4 Linux Security, Logging, and Process Management

90 min

Understand security configurations, analyze system log files, and monitor processes on Ubuntu.

LESSONS

- **Lesson 1:** Host Protection with the Uncomplicated Firewall (UFW)
- **Lesson 2:** Process Administration and System Monitoring
- **Lesson 3:** Log Analysis and System Troubleshooting via Journald

LAB EXERCISES

Securing Services and Analyzing Log Events on Ubuntu

AFTER COMPLETING THIS MODULE, YOU WILL BE ABLE TO

Configure host-level firewall protections with the Uncomplicated Firewall (UFW)

Monitor, manage, and scale system execution processes via terminal tools

Analyze central administrative system logs to debug platform failures

Identify the core concepts and terminology used in Linux Administration Essentials